

SwipeAI & Blunderr Dating

End-to-End Cloud Deployment, Cost Breakdown & Operations Manual

BACKEND SERVICE	MOBILE CLIENT	RECOMMENDED HOST	TOTAL MONTHLY COST
Spring Boot 3 + Java 21 (Virtual Threads)	React Native (Expo SDK 57)	Hetzner Cloud VPS (CX22) + Coolify	~€3.79 / month (~\$4.10 / ~₹350)

1. Executive Architecture Summary

SwipeAI is a production-engineered dating & social discovery platform consisting of a high-performance **Spring Boot 3** backend utilizing Java 21 Virtual Threads and an **Expo SDK 57** native mobile application. The platform features end-to-end encrypted (AES-256-GCM) chat lounges, masked WebRTC video calling (Virtual Chai), zero-screenshot app security, and database-level indexed geospatial search.

 Backend Architecture Spring Boot 3.3.x on Java 21 with Project Loom Virtual Threads, HikariCP connection pooling, Spring Data JPA with PostgreSQL index pushdown, STOMP over WebSockets for real-time messaging, and Cloudflare R2 media storage.	 Mobile Architecture Universal Expo SDK 57 React Native application with Expo Router, on-device AES-GCM cryptography (zero server plaintext knowledge), native FLAG_SECURE window protection, and Expo Push notification listeners.	 Real-Time & Media Persistent bi-directional WebSocket session layer for instant chat delivery and WebRTC SFU integration (via LiveKit) for masked in-app video and audio speed dating sessions.
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2. Cloud Platform Cost & Feasibility Matrix

A comprehensive evaluation of hosting options was conducted for the Spring Boot JVM application, PostgreSQL database, and persistent WebSocket connection requirements.

PLATFORM	CONFIGURATION	WEBSOCKETS	POSTGRESQL	MONTHLY COST (USD)	APPROX (INR)	DEVOPS COMPLEXITY
Hetzner Cloud (CX22) BEST VALUE	2 vCPU, 4 GB RAM, 40 GB NVMe	Native	Included (Self-Hosted)	\$4.10 / mo (€3.79)	~₹350 / mo	Low (with Coolify)
Railway (Hobby Tier)	1 GB RAM App + Managed DB	Native	1-Click Managed	\$5.00 – \$15.00 / mo	~₹420 – ₹1,250	Zero (Git Push)
Render (Starter + DB)	512 MB App + 256 MB DB	Native	1-Click Managed	\$14.00 / mo	~₹1,180 / mo	Zero (Git Push)

PLATFORM	CONFIGURATION	WEBSOCKETS	POSTGRESQL	MONTHLY COST (USD)	APPROX (INR)	DEVOPS COMPLEXITY
Fly.io	1 GB Machine + Fly Postgres	Native	Attached Volume	\$8.00 – \$12.00 / mo	~₹670 – ₹1,000	Low (CLI)
DigitalOcean Droplet	1 vCPU, 2 GB RAM, 50 GB SSD	Native	Self-Hosted	\$12.00 / mo	~₹1,000 / mo	Moderate
Google Cloud (Cloud Run)	Min instances = 1 + Cloud SQL	HTTP/2 & WS	Cloud SQL (db-f1-micro)	\$20.00 – \$28.00 / mo	~₹1,700 – ₹2,350	Moderate
AWS (App Runner + RDS)	App Runner + db.t4g.micro	Via ALB	Amazon RDS Postgres	\$30.00 – \$40.00 / mo	~₹2,500 – ₹3,350	Complex

💡 Why Hetzner Cloud is the Clear #1 Winner for SwipeAI

Standard PaaS providers (Railway/Render) charge \$20–\$35/mo for 2 GB of RAM. Hetzner provides **4 GB RAM and 2 vCPUs for only €3.79/month**. Because Spring Boot 3 + Java 21 with Hikari connection pools and virtual threads operates optimally with 1.5–2 GB RAM, Hetzner provides enterprise-tier JVM headroom at 1/8th of cloud PaaS pricing.

3. Zero-Cost 3rd-Party Production Ecosystem

By strategically combining the free tiers of specialized cloud providers, the operational costs for media storage, video calling, and push notifications are reduced to **\$0 / month**:

SERVICE AREA	PROVIDER	FREE TIER ALLOWANCE	PRODUCTION COST	WHY CHOSEN FOR BLUNDERR
Media & Photo Storage	Cloudflare R2	10 GB Storage Free Forever	\$0.00	\$0 Egress Bandwidth Fees. Unlike AWS S3, downloading high-res profile photos incurs zero bandwidth cost.
Audio & Video Speed Dating	LiveKit Cloud	50 GB / mo + 10 Concurrent Users	\$0.00	Zero WebRTC server maintenance; globally distributed STUN/TURN relays solve mobile 5G NAT traversal.
Push Notifications	Expo Push + Firebase FCM	Unlimited Notifications	\$0.00	Native background push token registration and deep-link routing on iOS and Android.
Database (Alternative)	Supabase / Neon	500 MB – 1 GB Managed Postgres	\$0.00	Can be used if external managed DB is preferred over self-hosting on Hetzner.

4. Hetzner VPS Setup & Operations Blueprint

When provisioning a bare Linux server (Ubuntu 24.04 LTS on Hetzner CX22), the following operational controls must be implemented:

A. Network Firewall Hardening

Configure a Hetzner Cloud Firewall rule directly in the web console before starting the server:

- Inbound Allowed:** Port 22 (SSH), Port 80 (HTTP), Port 443 (HTTPS).
- Inbound Blocked:** All other ports. **Never expose Port 5432 (PostgreSQL) or Port 8080 (Spring Boot) publicly.**

B. Reverse Proxy & WebSocket Configuration (Nginx / Caddy / Coolify)

WebSockets require persistent HTTP upgrade headers to prevent timeouts during real-time chat sessions:

```
server {
    server_name api.yourdomain.com;

    location / {
        proxy_pass http://localhost:8080;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    location /ws-chat {
        proxy_pass http://localhost:8080/ws-chat;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "Upgrade";
        proxy_set_header Host $host;
        proxy_read_timeout 86400s; # 24-hour persistence for active lounges
        proxy_send_timeout 86400s;
    }
}
```

C. JVM Memory Allocation (4 GB RAM Budget)

To prevent the Linux OOM (Out Of Memory) killer from terminating Spring Boot, strictly partition server RAM:

- **Spring Boot JVM:** `-Xms512m -Xmx2048m -XX:+UseG1GC` (Max 2.0 GB)
- **PostgreSQL 16:** `shared_buffers = 512MB, work_mem = 16MB` (Max 1.0 GB)
- **Operating System, Nginx & Buffer Cache:** Remaining 1.0 GB

D. Automated Database Backups

Set a daily cron job to dump PostgreSQL and upload an encrypted snapshot to Cloudflare R2:

```
# /etc/cron.daily/pg_backup
0 3 * * * docker exec swipeai-db pg_dump -U postgres swipeai_prod | gzip | aws s3 cp - s3://blunderr-b
ackups/db-$(date +%F).sql.gz --endpoint-url https://<R2_ACCOUNT_ID>.r2.cloudflarestorage.com
```

5. The Recommended Solution: Hetzner + Coolify

Coolify is a self-hosted, open-source PaaS that turns any bare Ubuntu VPS into a private Railway/Heroku cloud with zero monthly license fees.

⚡ Instant 3-Step Deployment Workflow

1. Create a **Hetzner CX22 server** (€3.79/mo) running Ubuntu 24.04 LTS.
2. SSH into the server and run the official installer:

```
curl -fsSL https://cdn.coollabs.io/coolify/install.sh | bash
```

3. Open `http://your-server-ip:8000` in your browser. Add your GitHub repository (SuniiilSingh/SwipeAI).

What Coolify automates out of the box:

- ☒ **Automatic Git Push Deployments:** Pushing to `main` triggers a zero-downtime rebuild.
- ☒ **1-Click Managed PostgreSQL:** Creates a private-network database container with persistent SSD storage.
- ☒ **Automatic Free SSL:** Issues and auto-renews Let's Encrypt certificates for `api.yourdomain.com`.
- ☒ **Automatic WebSocket Forwarding:** Traefik handles HTTP-to-WebSocket upgrades natively.
- ☒ **Automated Database Backups:** Built-in daily backup scheduler directly to Cloudflare R2 / S3.

6. Mobile App (SwipeAIApp) Distribution & Build Plan

The mobile application is built with **React Native (Expo SDK 57)**. Unlike the backend, it compiles into native binaries (`.apk` / `.aab` and `.ipa`) installed on end-user devices.

PLATFORM	DEVELOPER FEE	TESTING PHASE	PRODUCTION STORE	BUILD METHOD
Android	\$25 (One-time, lifetime)	Direct APK sharing via WhatsApp / Drive (\$0)	Google Play Console	Local Mac build (<code>npx eas-cli build --local</code>)
iOS	\$99 / year	TestFlight beta (up to 10,000 users)	Apple App Store	Local Mac build via Xcode (\$0)

Over-The-Air (OTA) Instant Updates

Using **EAS Update**, any JavaScript, UI design, or styling changes can be deployed directly to users' phones instantly without submitting a new release for Google or Apple review:

```
# Push instant OTA update to all active users
npx eas update --branch production --message "Fix UI layout and add festive icons"
```

Cost: **Free for up to 1,000 Monthly Active Users (MAU).**

7. Production Environment Variables Reference

Set these variables in Coolify, Docker, or your production hosting environment:

ENVIRONMENT VARIABLE	SAMPLE VALUE	DESCRIPTION
SPRING_PROFILES_ACTIVE	prod	Activates application-prod.yaml
SPRING_DATASOURCE_URL	jdbc:postgresql://postgres:5432/swipeai_prod	Internal container / cloud DB connection URL
SPRING_DATASOURCE_USERNAME	postgres	Database username
SPRING_DATASOURCE_PASSWORD	<strong-secure-random-pw>	Database password
JWT_SECRET	<32-character-random-secret-key>	HMAC-SHA256 signing secret for Auth tokens
CHAT_ENCRYPTION_KEY	<32-byte-aes256-secret-key>	Server-side AES-GCM fallback encryption key
SERVER_PEPPIER	<high-entropy-salt-string>	Shadow Shield contact phone hash pepper
FEATURE_R2_STORAGE_ENABLED	true	Enables Cloudflare R2 S3 photo storage
R2_ACCOUNT_ID	<cloudflare-account-id>	Cloudflare R2 Account Identifier
R2_ACCESS_KEY_ID	<r2-access-key>	Cloudflare R2 Access Token
R2_SECRET_ACCESS_KEY	<r2-secret-key>	Cloudflare R2 Secret Token
FEATURE_LIVEKIT_ENABLED	true	Enables WebRTC audio & video speed dating
LIVEKIT_HOST	wss://<project>.livekit.cloud	LiveKit WebRTC Signaling Host
LIVEKIT_API_KEY	<livekit-key>	LiveKit API Access Key
LIVEKIT_API_SECRET	<livekit-secret>	LiveKit API Secret for WebRTC room tokens
EXPO_PUBLIC_API_URL	https://api.yourdomain.com	Base REST & WS endpoint configured in SwipeAIApp